

Problem

Design a problem to help other students better understand passive high-pass filters.

Step-by-step solution

Step 1 of 1

Design a problem to help other students better understand passive highpass filters.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

In a highpass RL filter with a cutoff frequency of 100 kHz, $L = 40$ mH. Find R .

Solution

$$\omega_c = \frac{R}{L} = 2\pi f_c$$

$$R = 2\pi f_c L = (2\pi)(10^5)(40 \times 10^{-3}) = \underline{\underline{25.13 \text{ k}\Omega}}$$